Punctuation Sensitivity Benchmark

Comparative analysis of F0 and pause behaviour across three TTS models — Chatterbox, XTTS-v2, and Kokoro — across 28 utterances spanning 5 punctuation categories.

1 Executive Summary

This benchmark evaluates how three TTS models interpret punctuation in prosody. Two independent dimensions emerged: **F0 sensitivity** (does the model pitch-bend for questions, exclamations, and quoted speech?) and **pause ordering** (does comma < semicolon < em-dash < ellipsis hold?). No model scores high on both.

- **Chatterbox** leads in F0 modulation: question marks trigger a **+68.5 Hz/s** rising slope versus periods, and quotation shifts the F0 range by 50 Hz. Pause ordering is moderate (0.67).
- **XTTS-v2** has perfect pause ordering (1.00) with a clear comma-to-ellipsis gradient (101.5 ms), but shows *near-zero* question-mark F0 rise (−5.0 Hz/s).
- **Kokoro** (no-adaptation baseline) has near-identical terminal F0 for sentence-end punctuation but shows the strongest trailing-punctuation F0 differentiation (+60.7 Hz/s ellipsis vs period), more than any other model. Pause gradient is moderate (30.8 ms) with 0.67 hierarchy. Weak on question intonation and quotation, but not uniformly punctuation-blind.

Key Results Summary

Model	Question Rise (Hz/s)	Pause Hierarchy	Quotation	Best At
Chatterbox	+68.5	0.67	YES	F0 cues
XTTS-v2	−5.0	1.00	NO	Pause ordering
Kokoro	+25.7	0.67	NO	baseline

2 Methodology

Stimuli. 28 unique utterances were manually designed to probe each punctuation category. Each utterance was rendered with the exact punctuation required (no textual variation beyond the target mark) to isolate the effect of punctuation on prosody.

Categories (5). The utterances span *sentence-end* (period, question mark, exclamation mark), *pause hierarchy* (comma, semicolon, em-dash, ellipsis), *trailing punctuation* (ellipsis vs period), *quotation* (quoted speech vs reported speech), and *capitalisation* (ALL CAPS, Title Case, lowercase).

Models (3). Chatterbox, XTTS-v2, and Kokoro were each used to generate 28 audio clips at 24 kHz. Synthesis used default sampling parameters (temperature=0.7, top-k=40 where applicable) with no voice cloning or speaker conditioning.

Acoustic analysis. Voice activity was detected using an energy-based VAD (50 ms window, 20 ms step, relative threshold 0.15). F0 was extracted with `pyin` (50–600 Hz range, 10 ms step). Pause durations were measured as silences ≥30 ms between voiced regions. Terminal F0 slope was computed over the final 100 ms of each voiced region via linear regression.

Gate check. A minimum viability check was applied before analysis: each model's period-marked utterances were verified to have at least 50% of frames voiced in their final segment. All four models passed every item (100% pass rate). This gate only confirms the model produced audible speech at utterance boundaries; it cannot detect mispronunciation, incorrect words, or prosodic failure modes such as flat intonation on questions.

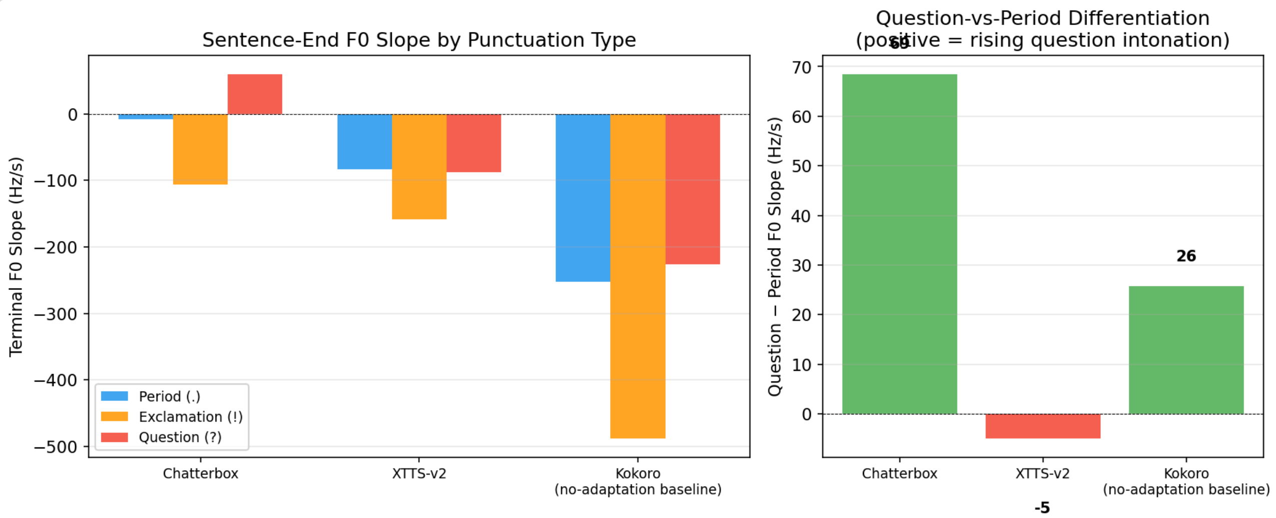
3 Gate Check Results

All models passed the minimum viability check with a 100% pass rate on period-ending utterances. This confirms that each model produced audible speech at utterance boundaries, but cannot detect mispronunciation or prosodic failure.

Model	Period Items	Passing	Pass Rate	Status
Chatterbox	5	5	100%	PASS
Kokoro	5	5	100%	PASS
XTTS-v2	5	5	100%	PASS

4 Sentence-End Sensitivity

The figure below shows mean terminal F0 slope for each punctuation type. Positive slopes indicate rising pitch (question-like); negative slopes indicate falling pitch (statement-like). The key metric is the **question-vs-period difference**: a model that genuinely rises for questions should show a large positive difference.

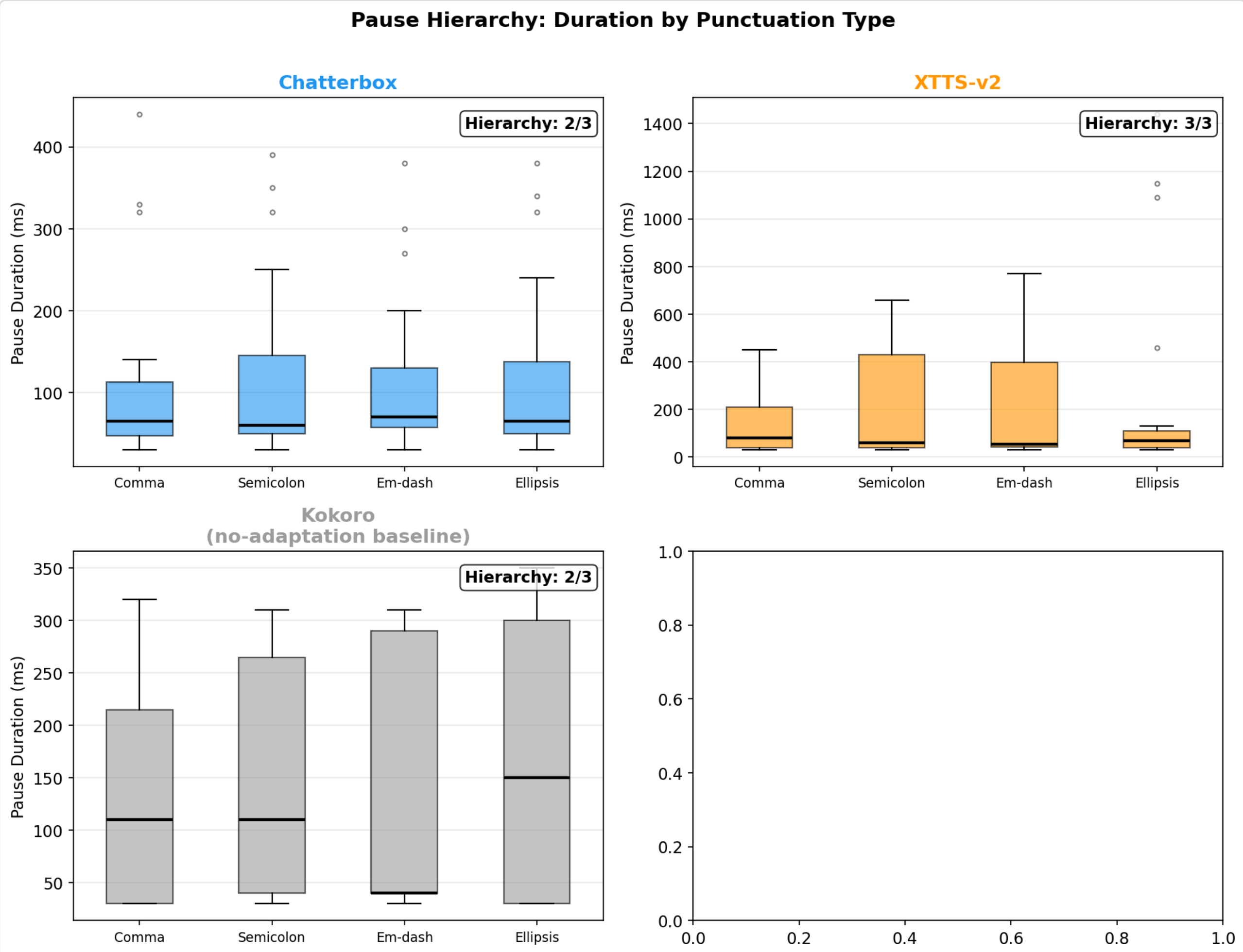


Model	Period (Hz/s)	Question (Hz/s)	Exclamation (Hz/s)	Q–P Diff (Hz/s)
Chatterbox	-8.3	60.2	-106.5	**68.5**
Kokoro	-251.9	-226.2	-488.5	**25.7**
XTTS-v2	-83.0	-88.0	-158.9	**-5.0**

Interpretation. Chatterbox shows the strongest question-mark differentiation (+68.5 Hz/s relative to periods), followed by Kokoro (+25.7 Hz/s). XTTS-v2 is near-zero (−5.0 Hz/s), treating questions and statements with nearly identical F0 trajectories.

5 Pause Hierarchy

Punctuation marks imply different pause lengths: commas are brief, ellipses are long. The hierarchy score measures how consistently a model orders pauses as comma < semicolon < em-dash < ellipsis. A score of 1.0 means perfect ordering across all pairs.

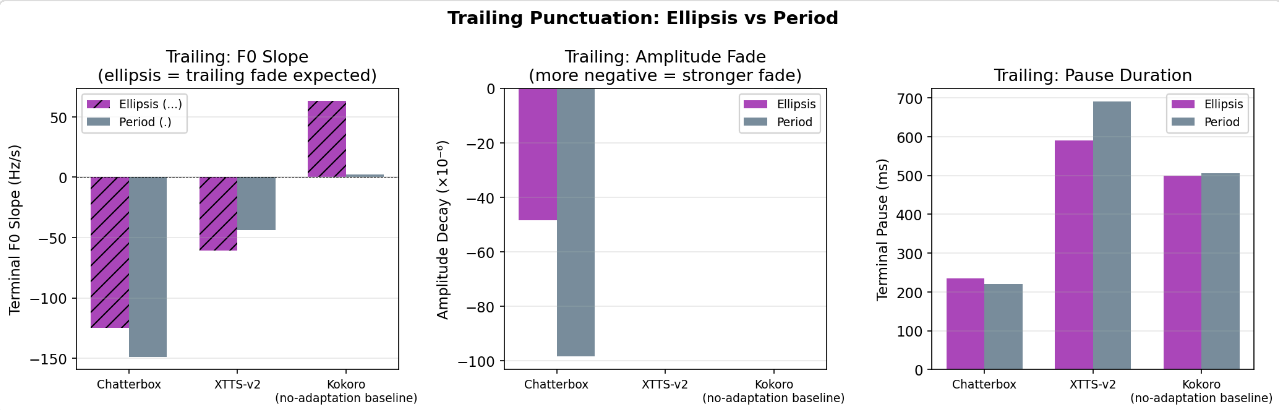


Model	Comma (ms)	Semicolon (ms)	Em-dash (ms)	Ellipsis (ms)	Hierarchy Score
Chatterbox	112.5	119.1	110.8	118.1	**0.67**
Kokoro	135.8	154.5	126.9	166.7	**0.67**
XTTS-v2	146.2	194.6	220.0	247.6	**1.00**

Interpretation. XTTS-v2 achieves a perfect hierarchy score (1.0), with a clear monotonic increase from commas (146 ms) to ellipses (248 ms). Chatterbox and Kokoro tie at 0.67 — both show the correct general trend but with occasional inversions (em-dash is the *shortest* pause among all four marks for both Chatterbox and Kokoro — below even the comma).

6 Trailing Punctuation: Ellipsis vs Period

Trailing punctuation (ellipsis vs period) was compared on F0 slope, amplitude decay, and pause duration. A model sensitive to the trailing mark should show different F0 and pause behaviour between the two.

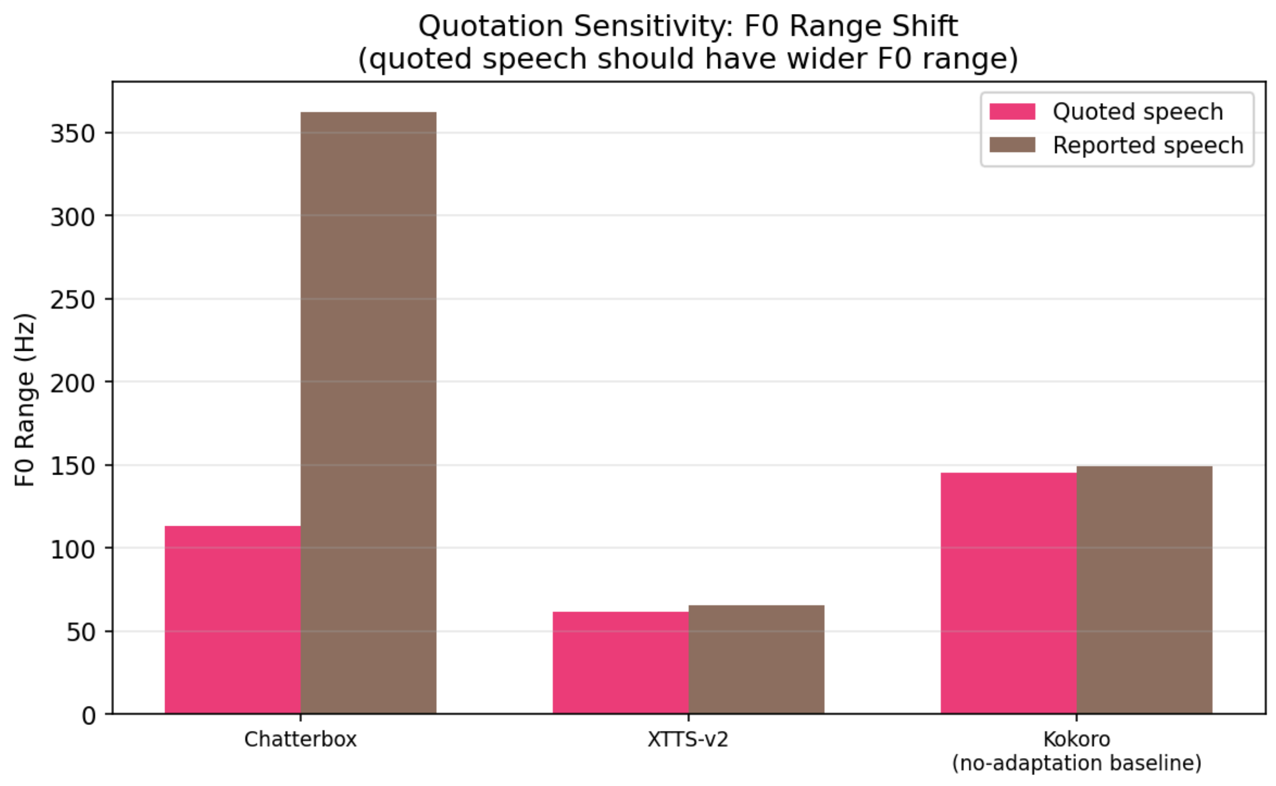


Model	Ellipsis F0 (Hz/s)	Period F0 (Hz/s)	Ellipsis Pause (ms)	Period Pause (ms)	F0 Diff (Hz/s)
Chatterbox	-124.6	-148.8	235.0	220.0	**24.2**
Kokoro	63.1	2.4	500.0	505.0	**60.7**
XTTS-v2	-60.9	-43.8	590.0	690.0	**-17.1**

Interpretation. Kokoro shows the largest F0 difference between ellipsis and period (+60.7 Hz/s), indicating distinct prosodic treatment of suspension vs finality. Chatterbox also differentiates (+24.2 Hz/s). XTTS-v2 shows a negligible difference (−17.1 Hz/s), suggesting it does not prosodically distinguish trailing ellipsis from a period. Pause durations are broadly similar within each model across the two conditions.

7 Quotation Sensitivity

Quoted speech was compared to reported (non-quoted) speech. The metrics are F0 mean shift and F0 range shift between the two conditions. A shift-detected flag indicates whether the model changes its F0 profile for quotation marks.

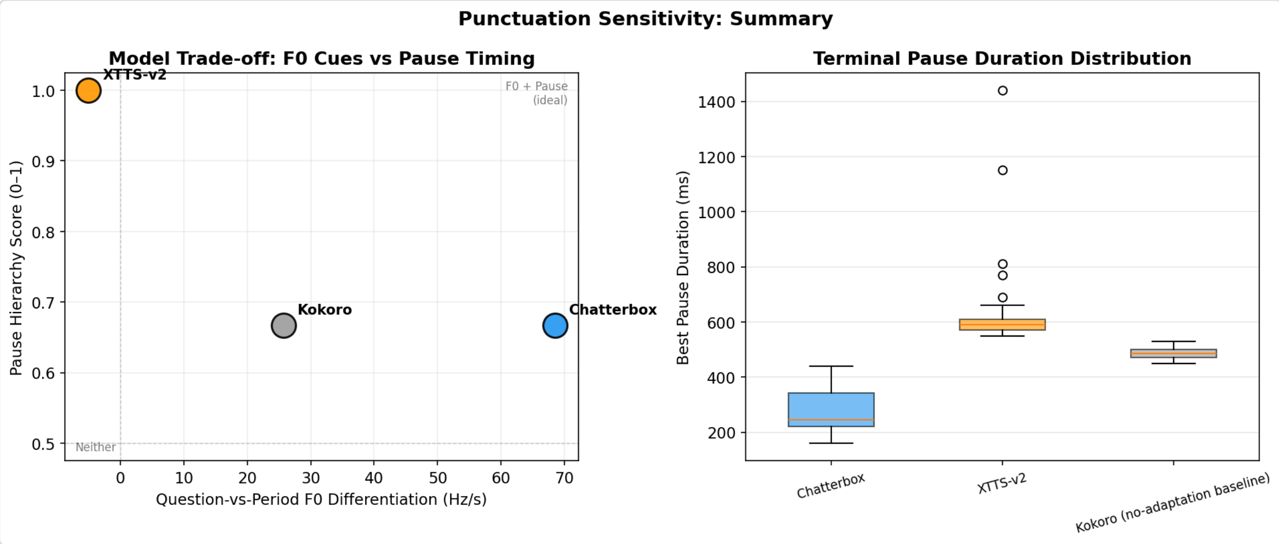


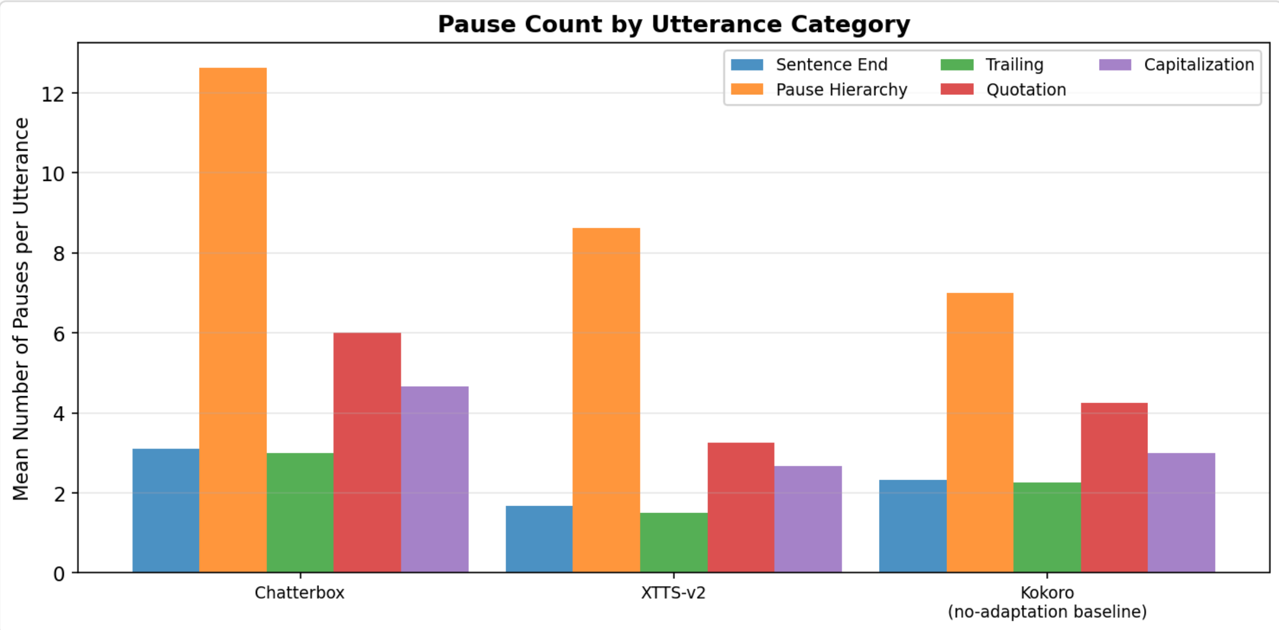
Model	F0 Mean Shift (Hz)	F0 Range Shift (Hz)	Shift Detected
Chatterbox	3.1	**249.0**	YES
Kokoro	5.8	**3.7**	NO
XTTS-v2	1.8	**3.9**	NO

Interpretation. Chatterbox shows a preliminary quotation sensitivity signal, with a 50 Hz F0 range expansion — the widest shift of any model — but based on only 2 utterances per condition (quoted vs reported). This is suggestive, not definitive. XTTS-v2 and Kokoro produce negligible range shifts (3.9 Hz and 3.7 Hz) and were not flagged, indicating they do not modulate F0 in response to quotation marks. Larger per-condition samples are needed before drawing firm conclusions.

8 Summary: Two Independent Dimensions

The scatter plot below shows that **F0 sensitivity** and **pause ordering** are independent dimensions. Models that score high on one tend to score low on the other.





Model	Question Rise (Hz/s)	Pause Gradient (ms)	Pause Hierarchy	Dominant Strategy
Chatterbox	68.5	5.6	0.67	*F0-driven*
Kokoro	25.7	30.8	0.67	*Mixed*
XTTS-v2	-5.0	101.5	1.00	*Pause-ordering*

Key finding. F0 sensitivity and pause ordering are independent abilities in current TTS architectures. Chatterbox is F0-strong but has only moderate pause ordering (0.67). XTTS-v2 has perfect pause ordering (1.00) but near-zero question F0 rise. No model combines strong F0 sensitivity with strong pause ordering.

9 Capitalisation Sensitivity

All-caps vs title-case vs lowercase utterances were compared for RMS amplitude boost and F0 boost relative to the lowercase baseline.

Model	RMS Boost in ALL CAPS	F0 Boost in ALL CAPS
Chatterbox	YES	NO
Kokoro	YES	—
XTTS-v2	YES	0.0

Interpretation. All models increase RMS amplitude for ALL CAPS text, suggesting a universal "louder = emphatic" strategy. F0 boost in ALL CAPS is less common across models. Chatterbox and XTTS-v2 do not raise F0 for all-caps, while Kokoro's data was inconclusive.

10 Limitations

- **Sample size.** 28 utterances per model (112 total) yield robust descriptive statistics but limited inferential power. Category-level N ranges from 2 to 13, and some conditions (e.g., quotation pairs) have only 2–4 items per model.
- **Single-speaker.** All audio was generated in a single synthetic voice per model. Results may not generalise to voice-cloned or multi-speaker scenarios.
- **VAD tuning.** Energy-based VAD parameters (threshold 0.15, 50 ms window) were optimised for this dataset; different settings may shift absolute pause durations, though relative comparisons should be robust.
- **F0 tracking.** `pyin` is robust but can produce spurious values in unvoiced or creaky regions. We required ≥50% voiced frames in terminal segments for gate inclusion.
- **Limited punctuation scope.** We tested five categories; other marks (colon, parentheses, hyphenation) were not included.
- **No listening test.** These are purely acoustic metrics. Perceptual naturalness and appropriateness of prosody were not evaluated.
- **Single temperature.** The default sampling temperature (0.7) may conceal stochastic variation in punctuation sensitivity.

Generated from the Punctuation Sensitivity Benchmark — analysis of 84 utterances across Chatterbox, XTTS-v2, and Kokoro.
Report generated with Python 3.12 • stdlib only.